Mathematics: analysis and approaches
Standard level
Paper 1
Topic 4 Statistics — cumulative frequency

5 November 2026

W4ss Academy practice set 2 | **Topic** Topic 4 Statistics — cumulative frequency | **Ref.** W4SS–MAA–QB–P1–02

Student name

20 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- You are **not** permitted access to any calculator for this paper.
- Section A: answer all questions. Answers must be written within the answer boxes provided.
- Section B: answer all questions on the answer pages at the end of this paper.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[19 marks]**.

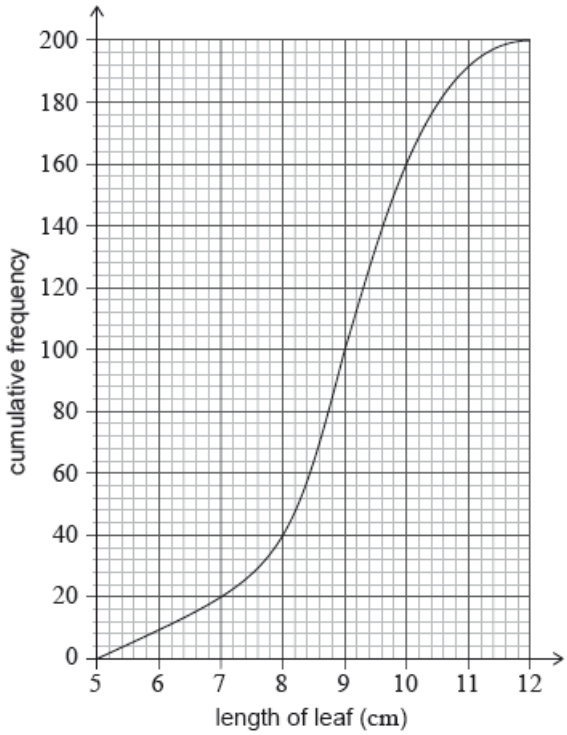
Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 2]

For a study, a researcher collected 200 leaves from oak trees. After measuring the lengths of the leaves, in cm, she produced the following cumulative frequency graph.

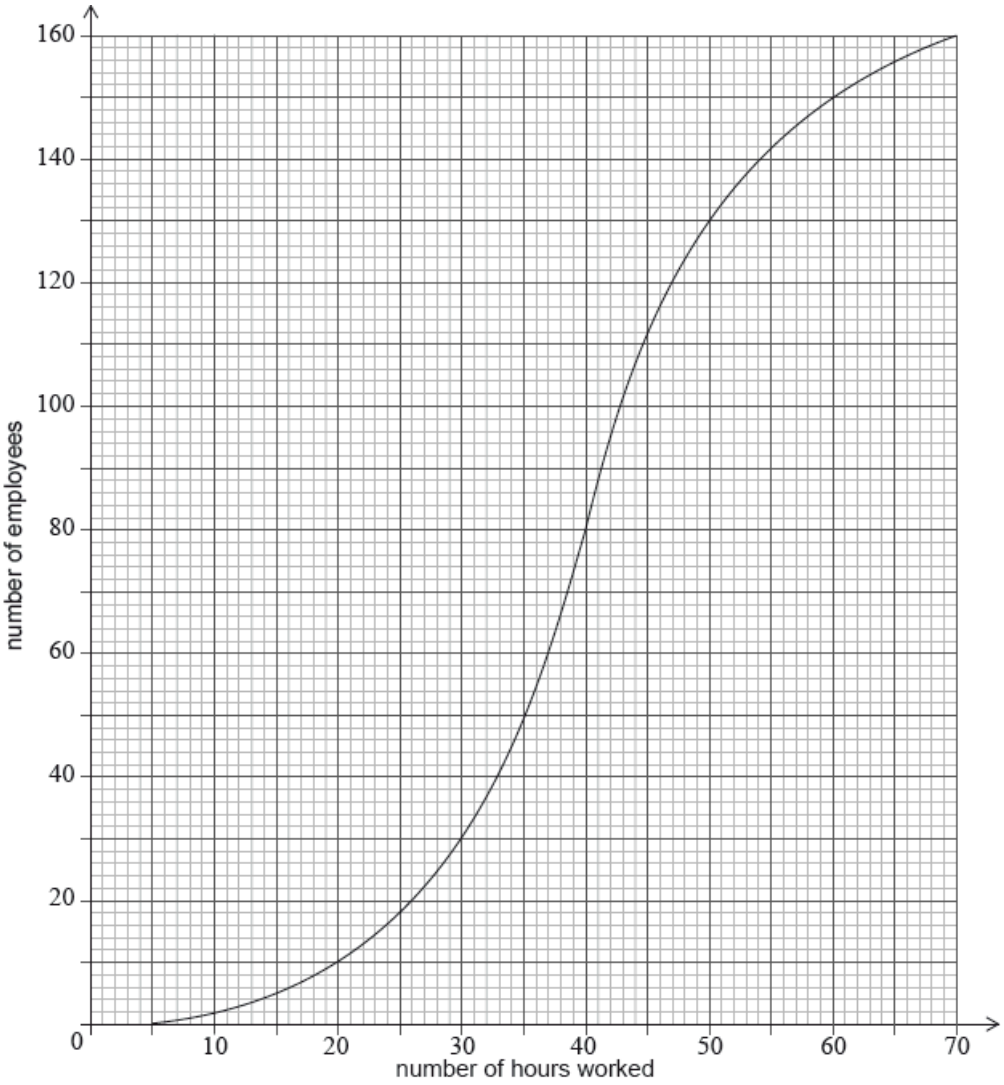


(a) Write down the median length of these leaves. [1]

(b) Write down the number of leaves with a length less than or equal to 8 cm. [1]

2. [Maximum mark: 2]

A city hired 160 employees to work at a festival. The following cumulative frequency curve shows the number of hours employees worked during the festival.



The city paid each of the employees £8 per hour for the first 40 hours worked, and £10 per hour for each hour they worked after the first 40 hours.

(a) Write down the number of employees who worked 50 hours or less. [1]

(b) Find the amount of money an employee earned for working 40 hours. [1]

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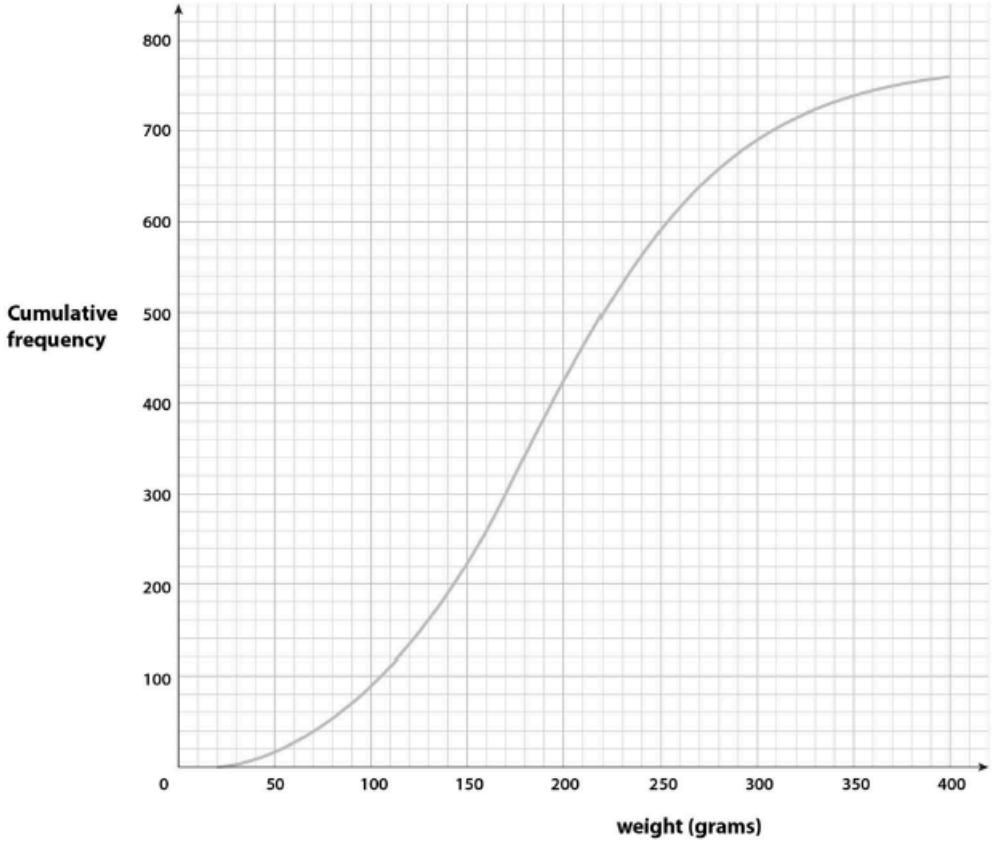
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3. [Maximum mark: 7]

A food scientist measures the weights of 760 potatoes taken from a single field and the distribution of the weights is shown by the cumulative frequency curve below.



(a) Find the number of potatoes in the sample with a weight of more than 200 grams. [2]

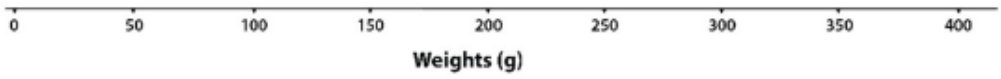
(b) (i) Find the median weight. [1]

(ii) Find the lower quartile. [1]

(iii) Find the upper quartile. [1]

The weight of the smallest potato in the sample is 20 grams and the weight of the largest is 400 grams.

(c) Use the scale shown below to draw a box and whisker diagram showing the distribution of the weights of the potatoes. You may assume there are no outliers.



[2]

(This question continues on the following page)

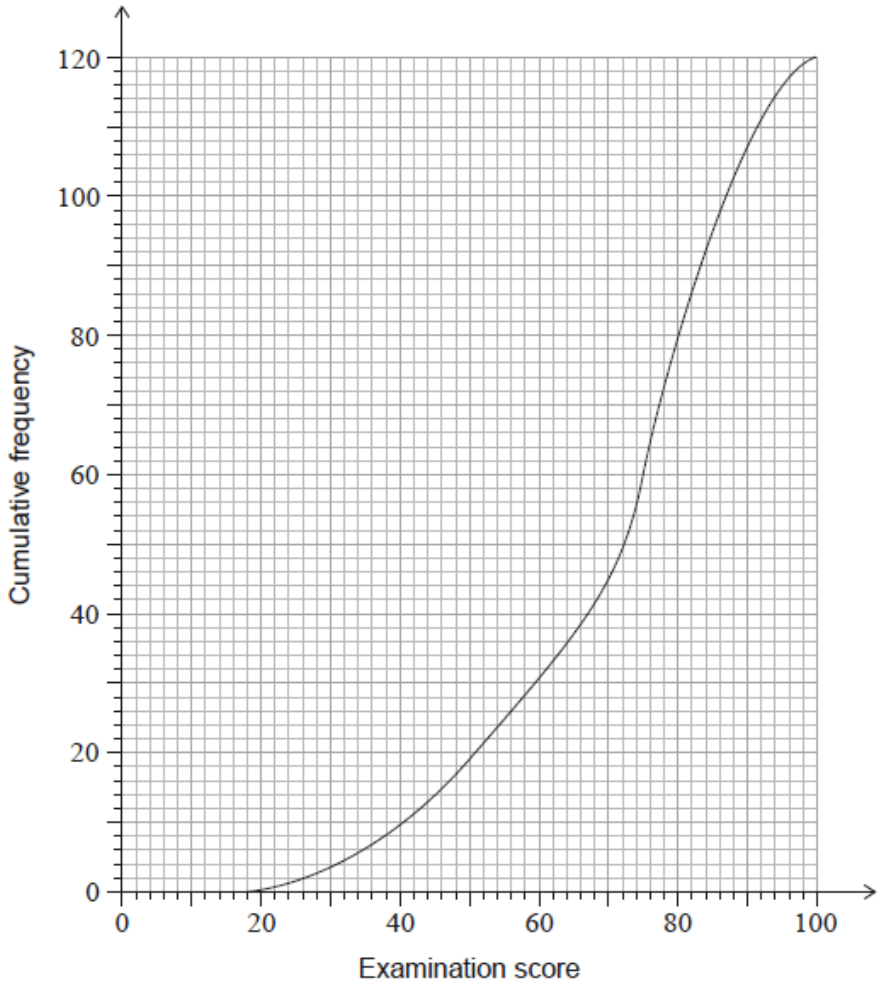
(Question 3 continued)

Section B

Answer **all** questions on the answer pages at the end of this paper. Please start each question on a new page.

4. [Maximum mark: 8]

A group of 120 students sat a history exam. The cumulative frequency graph shows the scores obtained by the students.



The students were awarded a grade from 1 to 5, depending on the score obtained in the exam. The number of students receiving each grade is shown in the following table.

Grade	1	2	3	4	5
Number of students	6	13	26	a	b

The mean grade for these students is 3.65.

(a) Find the median of the scores obtained. [1]

(b) Find an expression for a in terms of b. [2]

(This question continues on the following page)

Do not write solutions on this page.

(Question 4 continued)

(c) (i) Find the number of students who obtained a grade 5. [3]

(ii) Find the minimum score needed to obtain a grade 5. [2]

Do not write solutions on this page.

Section B — answers to question 4

Section B — answers to question 4 (continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 4 Statistics and probability), as chosen in the class QuestionBank test 2. Some of them come from former courses (Mathematics SL, Mathematical studies SL). They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	17M.1.SL.TZ2.T_6 (former Mathematical studies SL course) · 4.2 cumulative frequency graph, median
Section A — question 2	17M.1.SL.TZ2.S_8 (former Mathematics SL course) · 4.2 cumulative frequency curve — parts (a)(ii) and (b)(i) of the original, relabelled (a) and (b): the other parts are not in the test
Section A — question 3	EXN.1.SL.TZ0.4 · 4.2 cumulative frequency curve, quartiles, box-and-whisker diagram
Section B — question 4	21N.1.SL.TZ0.9 · 4.2 cumulative frequency graph · 4.3 mean of grouped data